

Sprout — Status & Product Roadmap

From hackathon MVP to a robust, research-grounded product

Prompt Wars Virtual Challenge 3 · "Create behaviour, not just awareness." · Internal briefing · 17 June 2026

1 · Executive summary

Sprout has moved well beyond the original "living world + nudge" MVP. It is now a **virtual-companion (creature) carbon platform**: you adopt and name a creature, perform **one verified high-impact green action a day**, and watch the creature grow, level up, and thrive — backed by streaks, collectibles, commitment pacts, a social grove (leaderboard), sponsor challenges, a community blog, and an AI coach + chat. The build is a client-first Next.js web app with a clean *data seam* that already anticipates a real backend.

~30

SOURCE MODULES

7

TEST SUITES (PURE LOGIC)

5

APP ROUTES / TABS

2

LLM PROVIDERS + FALLBACK

Where we stand: the *experience* is feature-complete and demo-ready; the *credibility & durability* layer is not. The social proof (leaderboard, friends, challenges, profile rank) and the activity verification are still **mock**. The single biggest leap from "MVP" to "product" is replacing those mocks with real accounts + backend behind the existing data seam, and replacing mock verification with real photo AI — without changing any component code.

2 · What is implemented today

2.1 Tech stack

Layer	Choice	Notes
Framework	Next.js 16.2.9 (App Router) · React 19	Web-first, the hackathon hard constraint
Language / styling	TypeScript · Tailwind v4	Custom "clay" soft-UI utility
State	Zustand v5 + <code>persist</code>	localStorage key <code>sprout-world-v2</code> , hydration-safe
Animation	Framer Motion v12	Reduced-motion respected throughout
AI	Gemini Flash (default) / Groq (swap) over REST	Server-only key; deterministic fallback
Testing	Vitest + Testing Library + jsdom	Pure-logic suites green
Data seam	<code>lib/data</code> interface	Mock today; Supabase/vision planned in-code

2.2 Feature inventory

Area	What's built	Status
Onboarding	Adopt + name a creature, pick a variant	DONE
Emotional core	Creature companion with moods/tints/levels; plus a rich health-driven Living World SVG (clouds, fireflies, smog haze, bloom/wilt bursts, optional sound)	DONE
Daily behaviour	Daily Ritual: one verified high-impact action/day with photo + AI verdict; heals the creature	DONE VERIFY=MOCK
Point-of-decision nudge	"Simulate a choice" sandbox: meal/commute/cooling/trip, options ordered low→high with carbon shown <i>before</i> choosing + impact badges	DONE
Footprint & insight	Totals, relatable equivalents (km / AC-hrs / ₹ / phone / tree-days), category breakdown	DONE
Normative social proof	"vs the average Indian" descriptive-norm bars	DONE
Gamification	Points, levels, savedKg, streaks, badges, collectibles + seed-pack variable reward	DONE
Commitment	Pacts (with a friend) — commitment device	DONE LOCAL
Social / collective	Grove leaderboard (global/friends/company), sponsor Challenges	MOCK DATA
Community	Blog (seeded + user posts)	DONE LOCAL
Profile	"You" tab: badges, rank summary	MOCK RANK
AI	Coach (one-shot next-best-action) + conversational CoachChat; never invents factors	DONE
Sharing	OG "world card" image route + Web-Share/clipboard snapshot	DONE
Transparency	Methodology page; hope-not-guilt framing	DONE
Accessibility	Live regions, role/aria, keyboard, non-color status, reduced-motion	STRONG

2.3 Information architecture

Five tabs: **World** (companion + daily ritual + coach), **Grove** (leaderboard), **Pacts** (challenges), **Blog**, **You** (profile). The decision-moment + footprint simulator is intentionally demoted to a "Curious? Simulate a choice" sandbox so the *companion* is the hero — the felt, emotional experience the brief rewards.

3 · How it honours the research thesis

The deep-research report identified three evidence-backed levers and a set of failure modes. Sprout maps onto all three:

Research lever / finding	How Sprout implements it
Emotional "living" visualization + loss aversion (Forest, Ant Forest)	The creature/world flourishes or wilts with behaviour; always <i>redeemable</i> (never permanent death) to avoid eco-guilt backlash
Point-of-decision nudges (JAMA: 23% shift)	Options ordered low→high with carbon shown before choosing; red "High impact" badge (negative framing on specific choices)
Social / collective accountability (OPower, dorm comps)	Grove leaderboard, friend pacts, sponsor challenges, normative "vs average" bars BACKEND PENDING
Translate CO₂ into relatable terms	km driven, AC-hours, ₹, phone charges, tree-days — India-flavoured
Few high-impact actions (Wynes & Nicholas)	One daily ritual, steered to diet/transport/energy/flights — not trivial actions
AI as translator/coach, not the product	Coach phrases pre-computed numbers; never invents factors; graceful fallback
Hope not guilt; honesty hook	"Built for behaviour and hope, not guilt"; methodology disclosure

4 · Honest gaps, risks & what could go wrong

RISK Mock social layer reads as "fake"

Leaderboard, friends, challenges and profile rank are static mock data. A sharp judge or real user will notice there are no real people. Credibility hit if presented as live.

RISK Mock verification = integrity gap

"Verified" green actions are a plausibility stub, not real photo AI. The whole "verified behaviour" promise rests on this. Must be clearly labelled until Phase C.

RISK No accounts / single-device

All state is localStorage. There is no auth, no cross-device sync, and a "leaderboard" cannot be genuinely social without real users.

RISK Client-clock streaks are spoofable

Streak/daily-care logic trusts the device clock; trivially gamed. Needs server-side time + anti-cheat for any real competition.

GAP Photos as data-URLs in localStorage

Bloats storage, no real upload/storage, lost on clear. Needs object storage.

GAP Test depth

Pure logic is tested; the new adopt → ritual → reward flows have no component/e2e coverage.

GAP Hollow-reward trap

Collectibles/points don't yet connect to real-world impact (the failure the research flagged: "points that lead nowhere").

GAP Retention mechanics

No recurring weekly feedback or "you vs your past self" — both strongly evidence-backed against novelty decay.

5 · Evaluation readiness (hackathon axes)

Axis	State	Action before submit
Web app	MET	—
Deployed link works (14-day manual eval)	PENDING	Deploy to Vercel; verify in incognito; client-first core means it can't break on backend sleep
GitHub repo + security	GOOD	Key server-only, <code>.env.example</code> present, <code>.env*</code> ignored — confirm no key in client bundle
AI-tool usage documented	PENDING	README narrative + LinkedIn post (tool choice, prompt flow, architecture)
Testing	PARTIAL	Add 1–2 component/e2e smoke tests
Accessibility	STRONG	Spot-check new tabs for keyboard + contrast
Problem-statement alignment	MET	understand / track / reduce / personalised insights all visibly present
Graceful degradation	MET	LLM fallback; no backend on critical path

6 · MVP → Product roadmap

Phase A — Harden & ship (submission-critical, days)

- Run `typecheck` + `lint` + `test:run` + `build` clean; fix any failures.
- **Transparently label all mock data** in-UI (a small "demo data" chip) so simulated social proof never reads as deceptive.
- Add component/e2e smoke tests for *adopt* → *daily ritual* → *reward* → *streak*.
- Deploy to Vercel; verify deployed link in incognito; confirm no API key in client bundle.
- Write README (architecture + AI-tool + prompt-flow narrative) and the LinkedIn post.

Phase B — Real backend & accounts (the biggest leap)

- Add **Supabase auth** + **Postgres**; implement `lib/data/supabase.ts` behind the existing `SproutData` seam — *components don't change*.
- Real leaderboard / friends / pacts / profile rank; invite codes for **hostel-floor** & **office-team** groups (maps to dorm-competition evidence).
- **Server-side time & streak validation** + rate-limits (anti-cheat) so competition is credible.
- Photo upload to Supabase Storage; Row-Level Security on all tables.
- Keep client-first fallback so the deployed link still degrades gracefully.

Phase C — Real verification & smarter AI

- Replace mock `verifyActivity` with **Gemini vision** photo verification (the integrity core).
- **RAG-ground the coach** on the curated India factor table to hard-stop hallucinated numbers.
- Personalised **weekly recap** (the recurring-feedback lever) + the app's own AI-footprint disclosure (honesty hook).

Phase D — Retention, scale & impact

- **Recurring weekly feedback + "you vs your past self"** trend (self-comparison $\approx 14\%$ reductions, sometimes beats neighbour comparison).
- **Real-world impact reward:** tie collectibles/points to a verified tree-planting partner so rewards aren't hollow (Ant Forest / Forest model; treat offsets as last-resort, not core).
- Co-op play (co-raise a creature, help a friend's streak) — Ant Forest's sustained-engagement mechanic.
- Accuracy upgrade: Climatiq/DEFRA-backed factors with disclosed vintage + uncertainty ranges; PWA + offline; Hindi/regional i18n; email/push nudges; retention analytics; B2B campus/office + verified-impact sponsor challenges (monetization).

7 · Research-backed changes to prioritise

Evidence	Concrete change in Sprout	Priority
Novelty decays in ~4–6 weeks; effects decay without recurring feedback (OPower re-spike)	Weekly creature "report card" + nudge; streak insurance for one missed day	High
Self-comparison ≈14% cuts, can beat neighbour comparison	Add "you vs last week" trend alongside "vs average"	High
Hollow rewards kill retention	Connect points to real verified impact (tree partner)	High
Self-reported ≠ measured behaviour (EAST/BIT)	Real photo/sensor verification; honest "estimated" labels until then	High
High-impact actions are concentrated	Bias the daily ritual + coach to diet/transport/energy/flights; show the multiplier vs trivial actions	Med
Greenwashing / individual-blame critique (BP origin)	Add a civic/systemic action category + transparent AI-footprint disclosure	Med
Ant Forest social play drives sustained use	Co-raise / help-a-friend mechanics on top of pacts	Med
Accuracy complaints sink incumbents	Disclosed factor vintage + uncertainty; Climatiq API in Phase D	Med

Open research questions to validate (post-launch)

- Does the **companion/pet** framing retain Indian hostel/office users better than a pure living-world? (A/B)
- Optimal **cadence** of recurring feedback before it becomes noise.
- The real high-impact action set for **growth-stage** users (first AC, first car, first flights).
- A verification UX that adds integrity *without* re-introducing tracking fatigue.

8 · Recommended next steps (in order)

1. **Ship Phase A:** green build, label mocks, smoke tests, deploy, README + LinkedIn. (unblocks submission)
2. **Supabase adapter (Phase B)** behind the data seam — real accounts + real social. (largest product jump)
3. **Self-comparison + weekly recap** — cheap, high-retention.
4. **Gemini-vision verification (Phase C)** — turns "verified" from a promise into a fact.
5. **RAG-grounded coach + AI-footprint disclosure** — accuracy + credibility.
6. **Real-impact rewards** — fixes the hollow-reward failure mode.

One rule to protect the score: the latest submission is the one that counts. Treat the final attempt as sacred — only re-submit after the deployed link is verified working and the build is green. Every new "product" feature should ship behind the data seam and keep the client-first fallback intact, so the judged link can never break.

9 · Verification & QA checklist

- `npm run typecheck` · `npm run lint` · `npm run test:run` · `npm run build` — all clean.
- Manual: adopt → daily ritual (with photo) → creature heals → reward opens → streak increments.
- Coach: with key (live AI) and with key unset (fallback still returns sensible coaching).
- A11y: keyboard-only traverse each tab; screen-reader announces world/creature state; reduced-motion honoured.
- Security: grep built client bundle for the API key (must be absent); `.env` ignored, `.env.example` committed.
- Deployed link opens cleanly in incognito with no console errors.

Appendix · India-localized factor anchors (disclosed vintage)

Factor	Value	Source
Grid electricity	0.71 kg CO ₂ /kWh (FY2024-25 weighted avg)	CEA CO ₂ Baseline Database
Food (per serving)	beef 15.5 · paneer 4.5 · chicken 1.8 · veg thali 0.7 kg	Our World in Data (Poore & Nemecek)
Transport (per km)	car 0.17 · auto 0.11 · bus/2W 0.05 · metro 0.02 · AC train 0.008 · flight 0.15	DEFRA / Climatiq / Indian Railways
Context anchor	avg Indian ≈ 2.0 tCO ₂ /yr (~5.48 kg/day) vs US ≈ 17.6	EDGAR / IEA / WRI

All numbers originate in `lib/carbon/factors.ts` and are surfaced on the in-app Methodology page. The AI layer only rephrases these values; it never computes or invents emission numbers.